PAPER ID-310590

Subject Code: BCE701

Roll No: | | | | | | | | | | | | | |

BTECH
(SEM VII) THEORY EXAMINATION 2025-26
ADVANCE STRUCTURAL DESIGN

TIME: 3 HRS **M.MARKS: 70**

Note: Attempt all Sections. In case of any missing data; choose suitably.
IS800:2007, IS3370, IS13920, Steel Table Allowed

SECTION A

1. Attempt *all* questions in brief. 2 x 07 = 14

Q no.	Question	CO
a.	List any two types of loads acting on a steel structure.	1
b.	Write the types of tension members.	2
c.	Write the assumptions made while designing a compression member,	2
d.	Define shape factor.	3
e.	Write the reason for high strength concrete is used in prestressed concrete.	4
f.	Give the name of joints in water tanks.	5
g.	What methods of design of water tanks?	5

SECTION B

2. Attempt any *three* of the following: 7 x 3 = 21

a.	List out advantages and disadvantages of steel structure.	1
b.	A single angle member carries a factored axial force of 400 kN. Design the sizing of single angle and sizing of lug angle. The yield strength and ultimate strength of the material is 250 MPa, and 410 MPA respectively.	2
c.	With neat sketch describe idealized strain curve for steel in tension.	3
d.	A prestressed concrete beam of rectangular x-section 120 mm wide and 300 mm deep is prestressed by 6 wires of 6 mm diameter, provided at an eccentricity of 55 mm. The initial stress in the wire is 1150 N/mm^2. Find the loss of stress due to creep of concrete. Find the loss of stress due to creep of concrete. Take E_s = 2 x 10^5 N/mm^2, E_c= 3x10^4 N/mm^2 and creep coefficient of concrete as 1.5.	4
e.	Explain with sketches various kinds of joints used in water tanks.	5

SECTION C

3. Attempt any *one* part of the following: 07 x 1 = 07

a.	A tie member of a roof truss consists of 2 ISA 100x75x8mm. The angles are connected to either side of a 10 mm gusset plates and the member is subjected to a working pull of 300 kN. Design the welded connection. Assume connections are made in the workshop.	1
b.	Write the classifications of bolted connection and types of bolts.	1

4. Attempt any *one* part of the following: 07 x 1 = 07

a.	How you will be finding design strength of compression member.	2
b.	Determine the design axial load on the column section ISMB450@710.3 N/m, height of column is 4 m and it is pin-ended. Assume that f_y=250 N/mm^2, f_u= 410 N/mm^2, E=2x10^5 N/mm^2.	2

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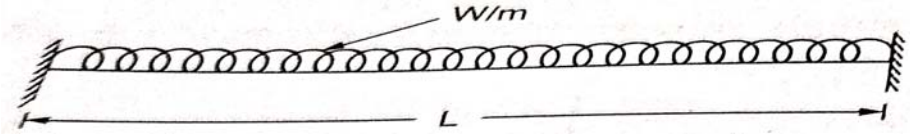
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5. Attempt any *one* part of the following: 07 x 1 = 07

a.	Find out shape factor for circular section of diameter 'd'.	3
b.	Determine collapse load for a fixed beam as shown in fig. 	3

6. Attempt any *one* part of the following: 07 x 1 = 07

a.	Write short notes on the following with neat sketch (i) Fressinet system of prestressing (ii) Magnel Blaton system of prestressing	4
b.	Determine the loss of prestress due to creep in concrete in a beam 200 mm x 400 mm, prestressed by a force of 500 kN by steel cables placed at eccentricity 50 mm. Take M45 concrete, creep coefficient =2	4

7. Attempt any *one* part of the following: 07 x 1 = 07

a.	Design a circular water tank with fixed base, resting on the ground, for a capacity of 500 kl. The depth of water in tank is 5 m and a freeboard of 200 mm is to be provided. Use M30 concrete and Fe415 steel. Use IS code method.	5
b.	Explain the various requirements regarding materials for design of water tanks.	5

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